Machine Learning Lab

Viva Questions & Answers

Experiments 1 to 11

Short, easy-to-revise answers for practical viva preparation

Included: experiment-wise questions, dataset names, key definitions, algorithm differences, evaluation metrics, and common viva questions.

Experiment 1: Basic Feature Engineering

Dataset: Titanic Passenger Survival Dataset

Q. What is feature engineering?

Answer: Preparing or creating useful input features from raw data to help a machine learning model.

Q. What is data preprocessing?

Answer: Cleaning and preparing data before training, such as handling missing values, encoding categories, and scaling features.

Q. What are missing values?

Answer: Empty or unavailable data entries. They can be removed or filled using mean, median, or mode.

Q. What is categorical data?

Answer: Data representing groups or labels, such as gender or passenger class.

Q. What is one-hot encoding?

Answer: It converts categories into separate binary columns containing 0s and 1s.

Q. What is feature scaling?

Answer: Bringing numerical features to comparable scales.

Q. What is the target in the Titanic dataset?

Answer: Survived, indicating whether a passenger survived.

Q. Why perform feature engineering?

Answer: To make data more useful and potentially improve model performance.

Experiment 2: PCA and LDA

Dataset: Wine Dataset

Q. What is dimensionality reduction?

Answer: Reducing the number of features while retaining useful information.

Q. What is PCA?

Answer: An unsupervised technique that creates fewer components capturing as much variance as possible.

Q. What is LDA?

Answer: A supervised technique that finds directions that help separate known classes.

Q. Main difference between PCA and LDA?

Answer: PCA focuses on variance without labels; LDA uses class labels to improve class separation.

Q. Is PCA supervised or unsupervised?

Answer: Unsupervised.

Q. Is LDA supervised or unsupervised?

Answer: Supervised.

Q. What is a principal component?

Answer: A new feature formed as a linear combination of original features.

Q. Why is dimensionality reduction useful?

Answer: It can reduce computation, simplify visualization, and remove redundant information.

Q. Why scale Wine features before PCA?

Answer: Features have different ranges, so scaling prevents large-scale features from dominating.

Experiment 3: Multiple Linear Regression

Dataset: California Housing

Q. What is regression?

Answer: Supervised learning used to predict continuous numerical values.

Q. What is Multiple Linear Regression?
Answer: It predicts a target using two or more input features.

Q. What is its equation?
Answer: y = b0 + b1*x1 + b2*x2 + ... + bn*xn.

Q. What is the target in California Housing?
Answer: Median house value.

Q. Simple vs multiple linear regression?
Answer: Simple uses one input feature; multiple uses two or more.

Q. What is train-test split?
Answer: Dividing data into training data for learning and test data for evaluation on unseen examples.

Q. What is MAE?
Answer: Mean Absolute Error: average absolute difference between actual and predicted values. Lower is better.

Q. What is R-squared?
Answer: A measure of how well the model explains target variation; closer to 1 generally indicates a better fit.

Q. Is regression supervised or unsupervised?
Answer: Supervised, because training data includes target values.

Experiment 4: Polynomial Regression

Dataset: Auto MPG

Q. What is Polynomial Regression?
Answer: It models nonlinear relationships by adding powers of input features, such as x^2 or x^3.

Q. Why use Polynomial Regression?
Answer: When a straight line does not adequately describe the relationship between input and output.

Q. What is a polynomial feature?
Answer: A feature created by raising an existing feature to a power or combining features.

Q. Is Polynomial Regression still a linear model?
Answer: It is nonlinear in inputs but linear in its coefficients.

Q. What is underfitting?
Answer: When a model is too simple to capture important patterns.

Q. What is overfitting?
Answer: When a model learns training data too closely and performs poorly on unseen data.

Q. What if polynomial degree is too high?
Answer: The model may become too complex and overfit.

Q. What is the target in Auto MPG?
Answer: Miles per gallon (MPG), representing fuel efficiency.

Q. How evaluate a regression model?
Answer: Using MAE, MSE, RMSE, and R-squared.

Experiment 5: Ridge and Lasso Regression

Dataset: Diabetes Dataset

Q. What is regularization?
Answer: Adding a penalty to the loss function to discourage large coefficients and help reduce overfitting.

Q. What is Ridge Regression?
Answer: Regression using an L2 penalty based on squared coefficients.

Q. What is Lasso Regression?
Answer: Regression using an L1 penalty based on absolute coefficient values.

Q. Difference between Ridge and Lasso?

Answer: Ridge shrinks coefficients toward zero; Lasso can make some coefficients exactly zero.

Q. What is alpha?

Answer: The regularization strength. Larger alpha generally means stronger regularization.

Q. Why use regularization?

Answer: To control complexity, reduce overfitting, and improve generalization.

Q. Can Lasso perform feature selection?

Answer: Yes. It can set some coefficients to zero.

Q. Why scale features for Ridge and Lasso?

Answer: Different feature scales can cause regularization to affect coefficients unevenly.

Q. What is the target in the Diabetes dataset?

Answer: A quantitative measure of disease progression.

Experiment 6: Logistic Regression

Dataset: Breast Cancer Wisconsin Diagnostic Dataset

Q. What is Logistic Regression?

Answer: A supervised classification algorithm used to predict class probabilities.

Q. Why is it called regression if used for classification?

Answer: It converts a score into a probability, then uses it to classify.

Q. What is the sigmoid function?

Answer: It converts a score to a probability between 0 and 1: sigmoid(z) = 1 / (1 + e^(-z)).

Q. What are the classes in the Breast Cancer dataset?

Answer: Malignant and benign.

Q. Classification vs regression?

Answer: Classification predicts categories; regression predicts continuous values.

Q. What is a confusion matrix?

Answer: A table comparing actual classes with predicted classes.

Q. What is accuracy?

Answer: The proportion of correct predictions: (TP + TN) / (TP + TN + FP + FN).

Q. What is precision?

Answer: Of examples predicted positive, the proportion that are actually positive.

Q. What is recall?

Answer: Of actual positive examples, the proportion correctly identified.

Q. Why is recall important in cancer detection?

Answer: Low recall can mean malignant cases are incorrectly classified as benign.

Experiment 7: Support Vector Machine (SVM)

Dataset used in our code: Breast Cancer Wisconsin Diagnostic Dataset

Q. What is SVM?

Answer: A supervised algorithm used for classification and regression.

Q. What is a hyperplane?

Answer: A decision boundary separating classes in feature space.

Q. What are support vectors?

Answer: Training points closest to the decision boundary; they help determine its position.

Q. What is the margin?

Answer: The distance between the decision boundary and the nearest data points.

Q. What is a kernel?

Answer: A function that lets SVM model different decision boundaries, including nonlinear ones.

Q. Name three SVM kernels.

Answer: Linear, polynomial, and RBF (Radial Basis Function).

Q. Linear vs RBF kernels?

Answer: Linear creates a linear boundary; RBF can model nonlinear boundaries.

Q. What is C?

Answer: It controls the trade-off between errors and margin width. Larger C penalizes errors more strongly.

Q. Why scale features before SVM?

Answer: SVM is sensitive to feature scales.

Q. Is SVM supervised or unsupervised?

Answer: Supervised.

Experiment 8: K-Means Clustering

Dataset: Mall Customer Segmentation

Q. What is clustering?

Answer: Grouping similar data points without predefined class labels.

Q. What is K-Means?

Answer: An unsupervised algorithm that groups data into K clusters around cluster centers.

Q. What does K represent?

Answer: The number of clusters.

Q. What is a centroid?

Answer: The center of a cluster, calculated as the mean of its data points.

Q. Main steps of K-Means?

Answer: Choose K; initialize centroids; assign points to nearest centroid; recalculate centroids; repeat until stable.

Q. Which features are used?

Answer: Annual Income and Spending Score.

Q. What is the elbow method?

Answer: A method to choose K by finding an elbow-shaped bend in the within-cluster squared-distance plot.

Q. What is inertia?

Answer: The sum of squared distances from points to their assigned centroids.

Q. Is K-Means supervised or unsupervised?

Answer: Unsupervised.

Q. One limitation of K-Means?

Answer: You must choose K beforehand; it can be sensitive to outliers and initialization.

Experiment 9: DBSCAN Clustering

Dataset: Synthetic Two Moons dataset

Q. What is DBSCAN?

Answer: An unsupervised algorithm that groups dense regions and identifies noise points.

Q. What does DBSCAN stand for?

Answer: Density-Based Spatial Clustering of Applications with Noise.

Q. What is eps?

Answer: The maximum distance for two points to be considered neighbors.

Q. What is min_samples?

Answer: Minimum points in an eps neighborhood for a point to qualify as a core point, including itself in scikit-learn.

Q. What is a core point?

Answer: A point with at least min_samples points in its eps neighborhood.

Q. What is a border point?

Answer: A non-core point within the neighborhood of a core point.

Q. What is a noise point?

Answer: A point that does not belong to any cluster under DBSCAN's density rules.

Q. K-Means vs DBSCAN?

Answer: K-Means requires K and uses centroids; DBSCAN uses density and can identify noise.

Q. What if eps is too small?

Answer: Many points may be marked as noise and clusters may split.

Q. What if eps is too large?

Answer: Different clusters may merge.

Q. Why use DBSCAN for Two Moons?

Answer: It can identify curved clusters that K-Means may struggle to separate.

Experiment 10: Ensemble Learning

Dataset: Breast Cancer Wisconsin Diagnostic Dataset

Q. What is ensemble learning?

Answer: Combining predictions from multiple models to improve performance or robustness.

Q. What is Random Forest?

Answer: An ensemble of decision trees whose predictions are combined, usually by voting for classification.

Q. What is Gradient Boosting?

Answer: An ensemble that builds models sequentially, with each new model working to reduce errors.

Q. Bagging vs boosting?

Answer: Bagging trains models independently on samples; boosting builds models sequentially.

Q. What is a decision tree?

Answer: A model that predicts by splitting data using feature-based rules.

Q. Why use Random Forest instead of one tree?

Answer: Combining trees often reduces variance and overfitting compared with a single deep tree.

Q. How does Gradient Boosting improve predictions?

Answer: It adds models sequentially to reduce a loss function.

Q. Which models were compared?

Answer: Logistic Regression, Random Forest, and Gradient Boosting.

Q. What is a hyperparameter?

Answer: A setting selected for training, such as number of trees or learning rate.

Q. How compare models?

Answer: Using accuracy, precision, recall, F1-score, and a confusion matrix.

Q. Does an ensemble always perform better?

Answer: No. It may cost more computation and may not beat a well-tuned individual model.

Experiment 11: Q-Learning

Environment: Custom 4 × 4 Grid World

Q. What is Reinforcement Learning?

Answer: An approach where an agent learns by interacting with an environment and receiving rewards or penalties.

Q. What is Q-Learning?

Answer: A model-free reinforcement learning algorithm that learns the value of taking an action in a state.

Q. What is an agent?

Answer: The learner or decision-maker interacting with the environment.

Q. What is an environment?

Answer: The world in which the agent takes actions and receives feedback.

Q. What is a state?

Answer: The current situation or position of the agent.

Q. What is an action?

Answer: A move the agent can take, such as up, down, left, or right.

Q. What is a reward?

Answer: Feedback indicating whether an action is beneficial or undesirable.

Q. What is a Q-table?

Answer: A table storing estimated values for each state-action pair.

Q. What is the Q-Learning update equation?

Answer: Q(s,a) <- Q(s,a) + alpha [r + gamma max Q(s',a') - Q(s,a)].

Q. What is learning rate alpha?

Answer: It controls how much new information changes the existing Q-value.

Q. What is discount factor gamma?

Answer: It controls how much future rewards are valued compared with immediate rewards.

Q. What is epsilon-greedy?

Answer: The agent explores with random actions with probability epsilon; otherwise it chooses the action with the highest current Q-value.

Q. Exploration vs exploitation?

Answer: Exploration tries new actions; exploitation chooses actions currently believed to be best.

Q. Is Q-Learning supervised or unsupervised?

Answer: Neither in the usual sense; it is reinforcement learning.

Q. Goal of the Grid World experiment?

Answer: Learn a policy that reaches the goal while maximizing cumulative reward and avoiding undesirable moves.

Common Viva Questions

Useful across all experiments

Q. What are the main types of machine learning?

Answer: Supervised learning uses labeled data; unsupervised learning finds patterns in unlabeled data; reinforcement learning learns through actions and rewards.

Q. Which experiments use supervised learning?

Answer: Experiments 3, 4, 5, 6, 7, and 10.

Q. Which experiments use unsupervised learning?

Answer: PCA in Experiment 2, K-Means in Experiment 8, and DBSCAN in Experiment 9. LDA in Experiment 2 is supervised.

Q. Which experiment uses reinforcement learning?

Answer: Experiment 11, Q-Learning.

Q. What is overfitting?

Answer: When a model performs well on training data but poorly on unseen data.

Q. What is underfitting?

Answer: When a model is too simple to learn important patterns.

Q. Why split data into training and testing sets?

Answer: To evaluate performance on data the model has not seen during training.

Q. What is a feature?

Answer: An input variable used by a model to make predictions or discover patterns.

Q. What is a label or target?

Answer: The output a supervised model is trained to predict.

Q. What is model evaluation?

Answer: Measuring model performance using suitable metrics.